Basic Technical Data

nominal electrical output			528	kW
maximum heat output [1)]			643	kW

load	50	75	100	%
maximum heat output[1)]	364	498	643	kW
fuel input	703	997	1306	kW
electrical efficiency	37,5	39,7	40,4	%
heat efficiency	51,8	50,0	49,2	%
total efficiency (fuel utilization)	89,3	89,7	89,6	%
gas consumption	74,4	106	138	m^3/h

The Basic Technical Data are applicable for the standard conditions pursuant to the "Technical instructions" document.
The minimum permanent electrical output must not drop below 50 % of the nominal output.
Gas consumption is expressed under the invoicing conditions (15°C, 101.325 kPa)
Gas consumption tolerance, or fuel input tolerance, at 100% load is +5%.
Tolerances of other parameters are mentioned in "Technical Instructions-Validity of Technical Data" document.
1) Maximum heat output is a sum of heat outputs of secondary circuit with exhaust gas cooled to 120°C and aftercooler circuit

Observance of Emission Limits

emissions	NOx	CO	
with 5% of O_2 in exhaust gases	500	300	mg/Nm^3

Generator

used types	LSA 49.3 M8	
producer	LEROY SOMER	
cos φ	1,0	
efficiency in the working point	96,3	%
voltage	400	V
frequency	50	Hz

Engine

type	E 3262 LE 202	
producer	MAN	
number of cylinders	12	
arrangement of cylinders	V	
bore × stroke	132/157	mm
displacement	25800	cm^3
compression ratio	12 : 1	
speed	1500	rpm
oil consumption, normal / max.	0,2/0,3	g/kWh
max. engine output	550	kW

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Thermal System

Secondary circuit

heat carrier	water	
circuit's heat output	601	kW
nominal water temperature, input / output	70/90	°C
nominal temperature drop	20	°C
return water temperature, min / max	40/70	°C
nominal flow rate	7,2	kg/s
max. working pressure	1600	kPa
water volume in CHP unit circuit	56	dm^3
pressure loss at the nominal flow rate	15	kPa

Utilization of exhaust gas output for other purposes

heat output of exhaust gases (cooling to 120°C)	251	kW
exhaust gas temperature	402	°C

Primary circuit

circuit's heat output	601	kW
max. working pressure	300	kPa
water volume in CHP unit circuit	315	dm^3

Aftercooler circuit

heat carrier	water + ethylene glycol	
ethylene glycol's concentration	35	%
circuit's heat output	42	kW
max coolant temperature at the input	42	°C
nominal flow rate	2,5	kg/s
pressure reserve at the nominal flow rate	70	kPa
max. working pressure	250	kPa
water volume in CHP unit circuit	10	dm^3

Fuel gas inlet

low heat value	34	MJ/m³
min. methane number	80	
gas pressure	2 ÷ 10	kPa
max. pressure change under varying consumption	10	%
max. gas temperature	35	°C

Combustion and Ventilation Air

unused heat removed by the ventilation air	62	kW
aspirated air temperature, min / max	10/35	°C
amount of combustion air	2062	Nm³/h
max. amount of ventilation air at the outlet flange	15700	m³/h
max. air temperature at the outlet flange	50	°C
max. counter-pressure at the ventilation air offtake flange [1)]	95	Pa
max. counter-pressure at the ventilation air flanges in Super Silent version [2)]	50	Pa

1) Valid for standard noise parameters
2) The sound protection version Silent or Super Silent is not included in the standard scope of delivery but it can be ordered.

Exhaust Gas and Condensate Outlet

amount of exhaust gases	2142	Nm³/h
exhaust gas temperature, nominal / max	120/150	°C
max. back-pressure of exhaust gases downstream the CHP unit flange [1)]	20	mbar
pressure loss of the freely delivered silencer	10	mbar
permissible pressure loss of the interconnecting exhaust piping	10	mbar
speed of exhaust gases at the outlet (DN 200)	27,3	m/s

1) Valid for standard version (without economizer)

Lubricant Charges

amount of lubrication oil in the engine	90	dm³
replenishment oil tank volume	130	dm³

Noise Parameters

	Standard	Super Silent	
sound enclosure of CHP unit at 1m	78	65	dB(A)
ventilation outlet of sound enclosure at 1m	94	65	dB(A)
exhaust gases outlet at 1m from the silencer flange	65	65	dB(A)

Electrical Parameters

nominal voltage	230/400	V
nominal frequency	50	Hz
power factor [1)]	0,8	
nominal current at cos φ=0.8	952	A
generator circuit breaker	NS1000HFE2	
short-circuit resistance of switchboard	35	kA
contribution of the actual source to the short-circuit current	< 10	kA
protection of switchboard's power part closed/open	IP 31/00	
protection of switchboard's control part closed/open	IP 31/00	
recommended superior protection	1000	A
recommended connection cable [2)] (length< 50m, at t<35°C)	3× (NYY-J 3×150+70)	

1) Power factor adjustable from 0,8C ÷ 1 ÷ 0,8L (range from 0.8C ÷ 1 must be verified according to the various types of generators).
L = inductive load - overexcited
C = capacitive load - underexcited
Operation of the generator with a power factor of less than 0.95 causes a power limitation sets the following table:

power factor [-]	1	0,95	0,8
output [% Pnom]	100	100	98

2) The stated cables are for information only. A check calculation for temperature rise and voltage drop must be made according to the actual length, placement and type of the cable (maximum allowed voltage drop is 10 V).

Colour Version

sound enclosure, base frame	RAL 5015 (blue)

Unit Dimensions and Weights*

length, total	4950	mm
width	1800	mm
total height	2350	mm
service weight of the entire CHP unit	9800	kg

** approximate values*

Caution

Manufacturer reserves the right to alter this document and the linked source materials.